Evolution- New Synthesis & Mechanisms

Some Historical Perspective….

Creationism

All living things are made by God

Only God can make new forms

Origin of life

Diversity of life

~ 2 million life forms (species)

Some early philosophers and naturalists…

Maupertuis

Diderot

Darwin was not the first to question the idea that species were fixed, although it was certainly the commonly held belief at his time.

The French naturalist Jean-Baptiste Lamarck put forward the idea that species changed over time in 1809 in his book *Philosphie Zoologique*.

Lamarck suggested the idea of **transformism**. He **did not** suggest that lineages branched and went extinct.

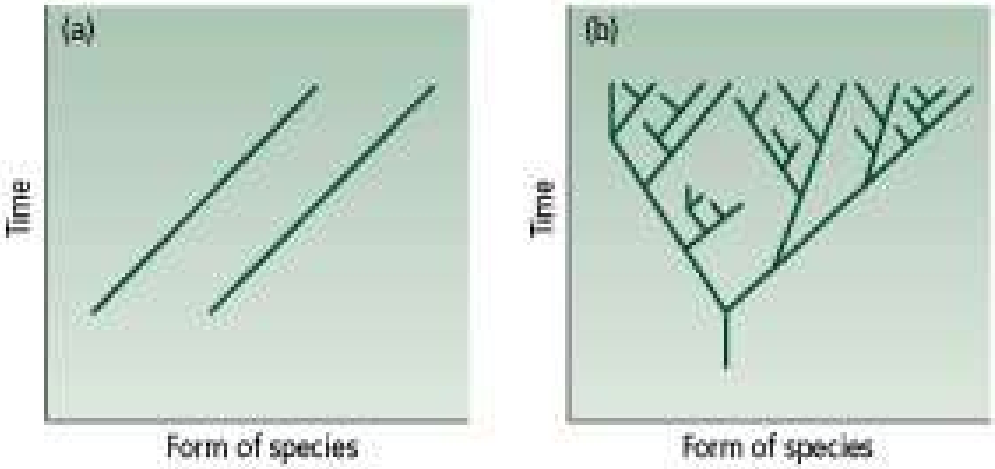


Figure 1.2
(a) Lamarckian "transformism," which differs in two crucial respects from evolution as Darwin imagined it. (b) Darwinian evolution is tree-like, as lineages split, and allows for extinction.

Lamarck's Theory of Evolution 1809

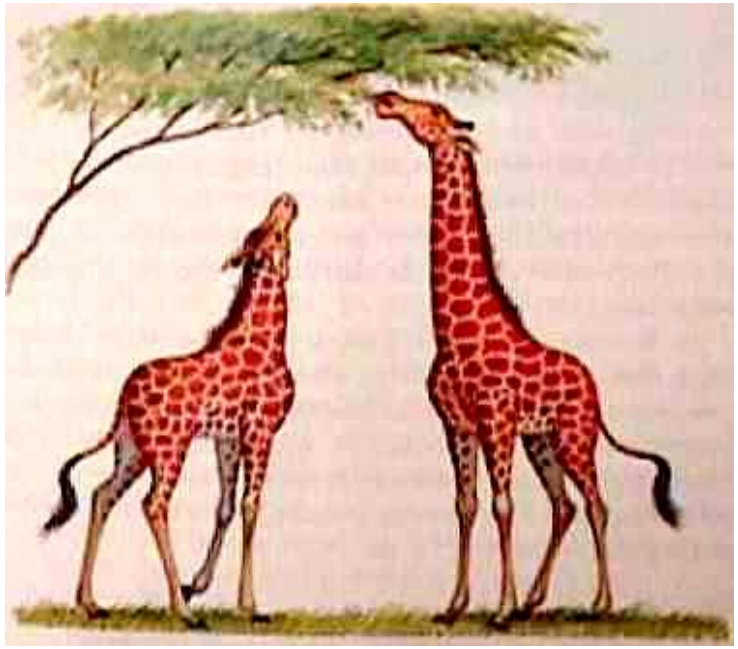
Lamarck's Theory actually had two parts:

1. Transformism Use and Disuse

2. Inheritance of Acquired Characteristics.

Organisms have a drive to become complex

Organisms adapt to their environment



Charles Darwin

1809-1882

"As far as I can judge of myself I worked to the utmost during the voyage from the mere pleasure of investigation, and from my strong desire to add a few facts to the great mass of facts in natural science."
-- Charles Darwin

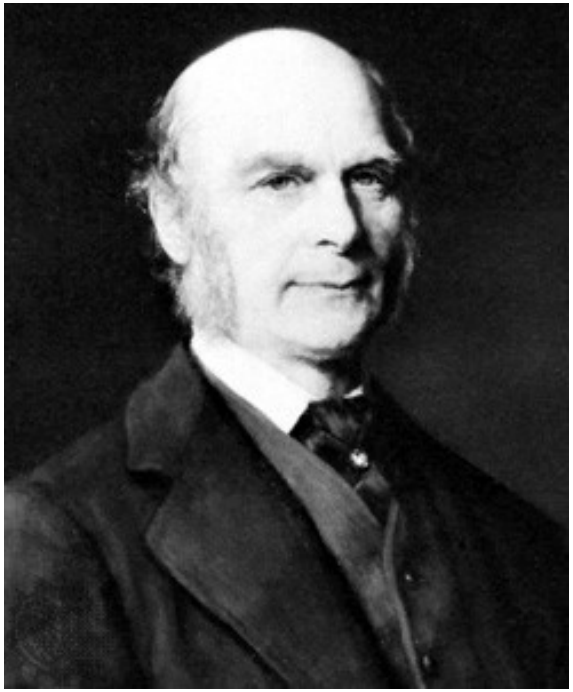
“I happened to read for amusement Malthus on Population and, being well prepared to appreciate the struggle for existence which everywhere goes on from long-continued observation of the habits of animals and plants, it at once struck me that under these circumstances favourable variations would tend to be preserved and unfavourable ones to be destroyed.”

Darwin, on reading Malthus’ Essay on the Principle of Population in 1838.

Thomas Malthus

Darwinism

"Species are only strongly marked varieties with the intermediate gradations lost." – *Origin of Species*



Sir Francis Galton (1822 – 1911)

Galton's law of regression: Parents who deviated from the average of the population tend to produce offspring closer to the average.

Discontinuous variation leads to the origin of new species.

Discontinuous variations are sudden, unpredictable changes in gene or chromosome mutations that can cause qualitative traits to jump from one category to the next. For example, most birds of a species can only be one of a few different colors

A more popular objection had to do with the role of chance, and the difficulty of making big jumps on the basis of chance and natural selection alone.

A comparative anatomist, George Mivart published a book in 1871 - he listed a number of structures that would be maladaptive in their early stages.

Wing, for example.

Some biologists tried to get around the difficulty by thinking of a process other than natural selection that could work in the early stages of a new structure or organ.

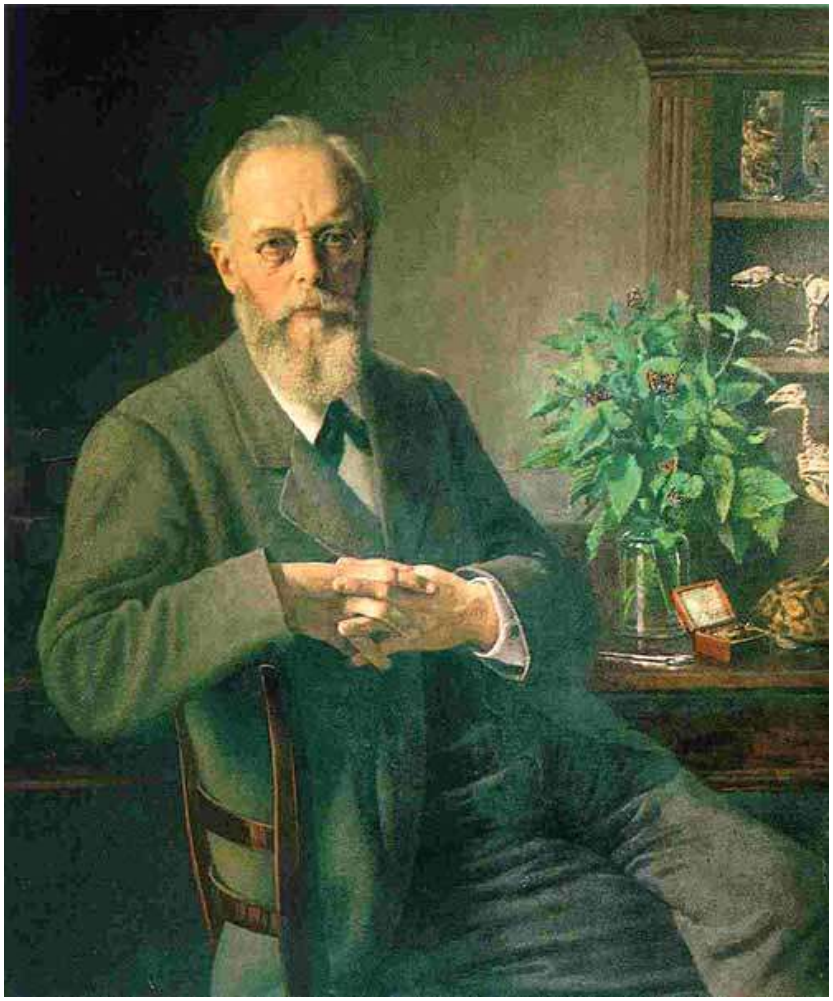
→ "directed mutation".

The idea is that offspring tended to differ from their parents in a particular direction.

In other words, early birds would tend to produce offspring with more developed wings, even though there was no advantage to it. Lamarckian.

In the 1880's, Lamarckian inheritance was largely debunked through the experiments of August Weismann.

His work led to a rediscovery of the work of….



….Gregor Mendel.

Early on, the effect of Mendel's work was to cast doubt on Darwin's theory.

The early Mendelian geneticists like Hugo de Vries and William Bateson opposed natural selection.

They thought that evolution proceeded in big jumps powered by macromutations. Big difference between parent and offspring.

On the other hand, another school of geneticists called themselves **biometricians**. They were led by Karl Pearson.

Biometricians looked at small differences between individuals instead of large. Source of much of biological statistics.

Saw evolution as more of a statistical shift of the whole population instead of the production of an entirely new type of organism by macromutation.

They tended to be more sympathetic to Darwin than the Mendelians.

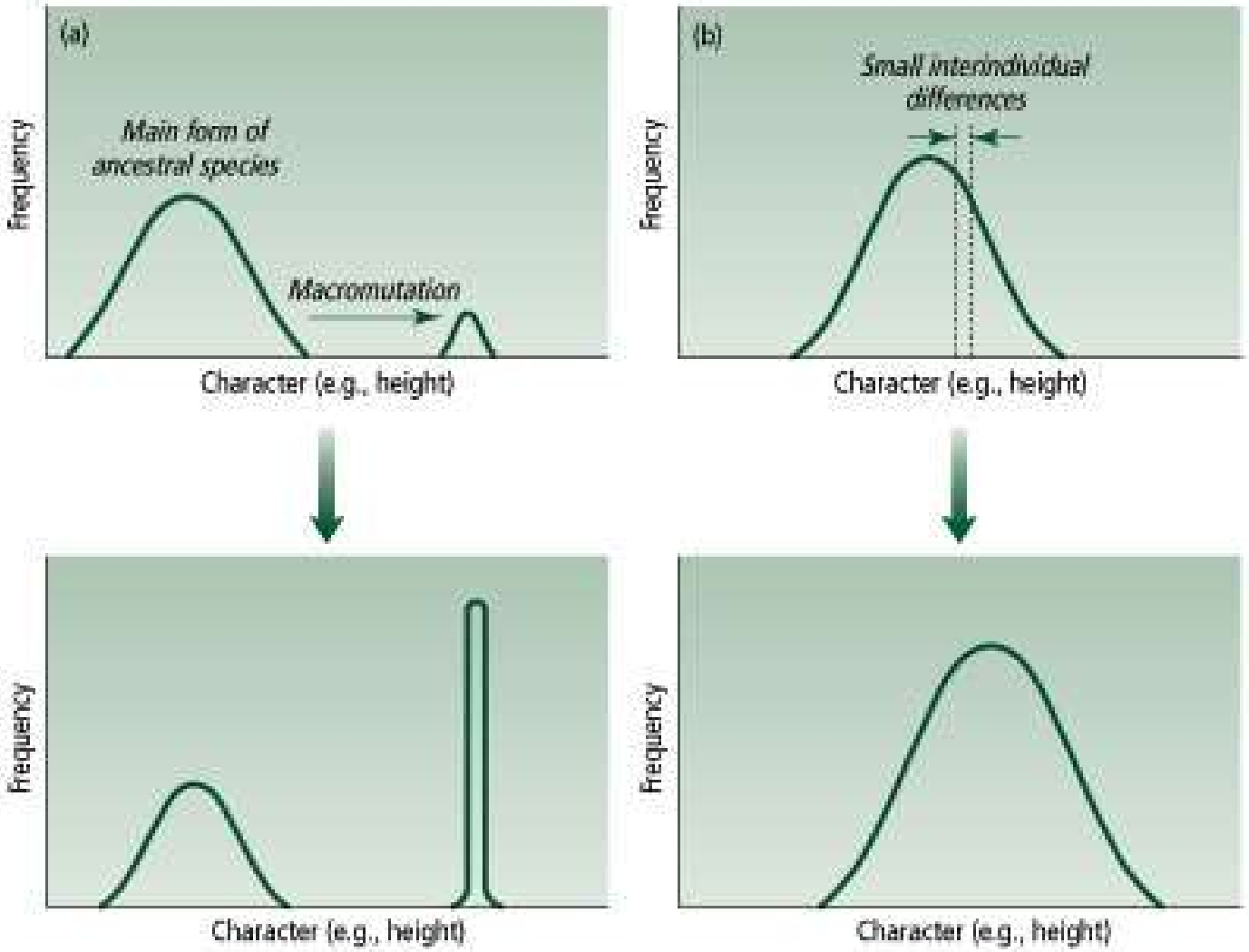
Darwinism – post Darwin, there were 2 ideas:

The **Biometricians** -the importance of continuous variation.

The **Mendelians** - most heritable characters were discontinuous.

Led to two schools
--the mutationists and
-- the selectionists.

Figure 1.7
Early Mendelians and biometricians. (a) Early Mendelians studied large differences between organisms, and thought that evolution happened when a new species evolved from a "macromutation" in its ancestor. (b) Biometricians studied small interindividual differences, and explained evolutionary change by the transition of whole populations. Mendelians were less interested in the reasons for small interindividual variations. The figure is a simplification — no historic debate between two groups of scientists lasting for three decades can be fully represented in a single diagrammatic contrast.



The Modern Synthesis

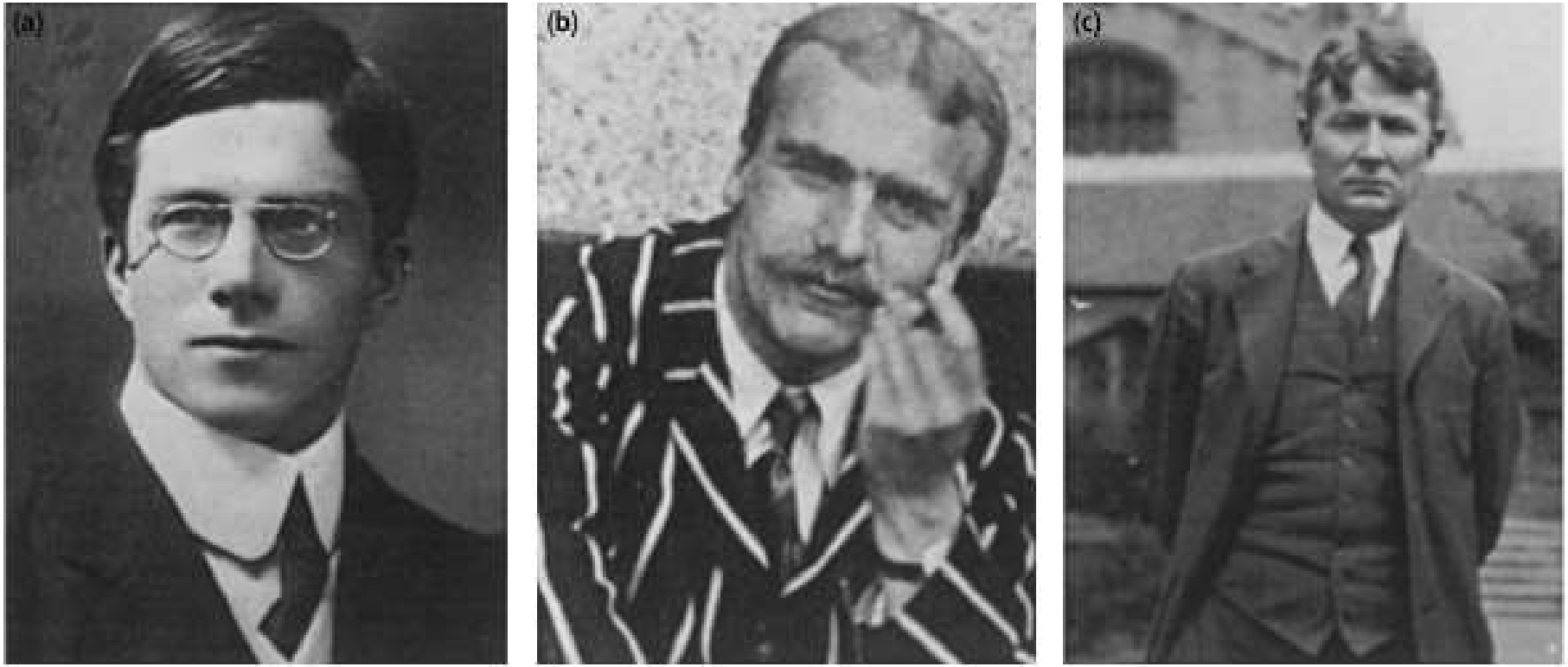


Figure 1.8
(a) Ronald Aylmer Fisher (1890–1962) in 1912, as a Steward at the First International Eugenics Conference.
(b) J.B.S. Haldane (1892–1964) in Oxford, UK in 1914.
(c) Sewall Wright (1889–1988) in 1928 at the University of Chicago.

Figure 1.9
Theodosius Dobzhansky (1900–75) in a group photo in Kiev in 1924; he is seated second from the left at the front, in the great boots.

Theodosius Dobzhansky immigrated from Russia in 1927 and began studies on evolution in fruit flies.

The question of how species originate was investigated by Ernst Mayr.

"Biological Species Concept."

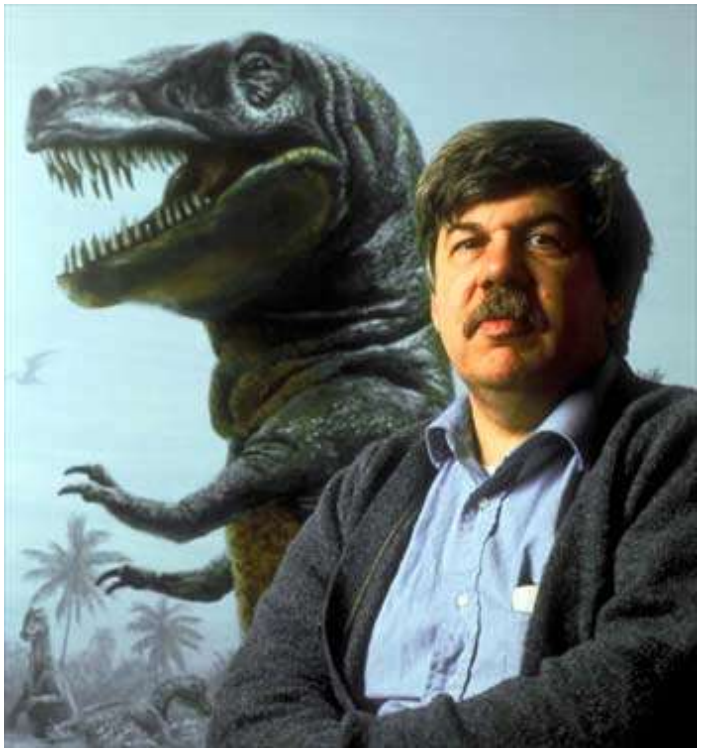
In his landmark 1942 book, Systematics and the Origin of Species, Mayr outlined the concept of “biological species.”

June 1928: Ernst Mayr with Sario, one of his Malay assistants, in the former Dutch New Guinea.

American paleontologist known for his contributions to evolutionary theory and to the understanding of intercontinental migrations of animal species in past geological times.

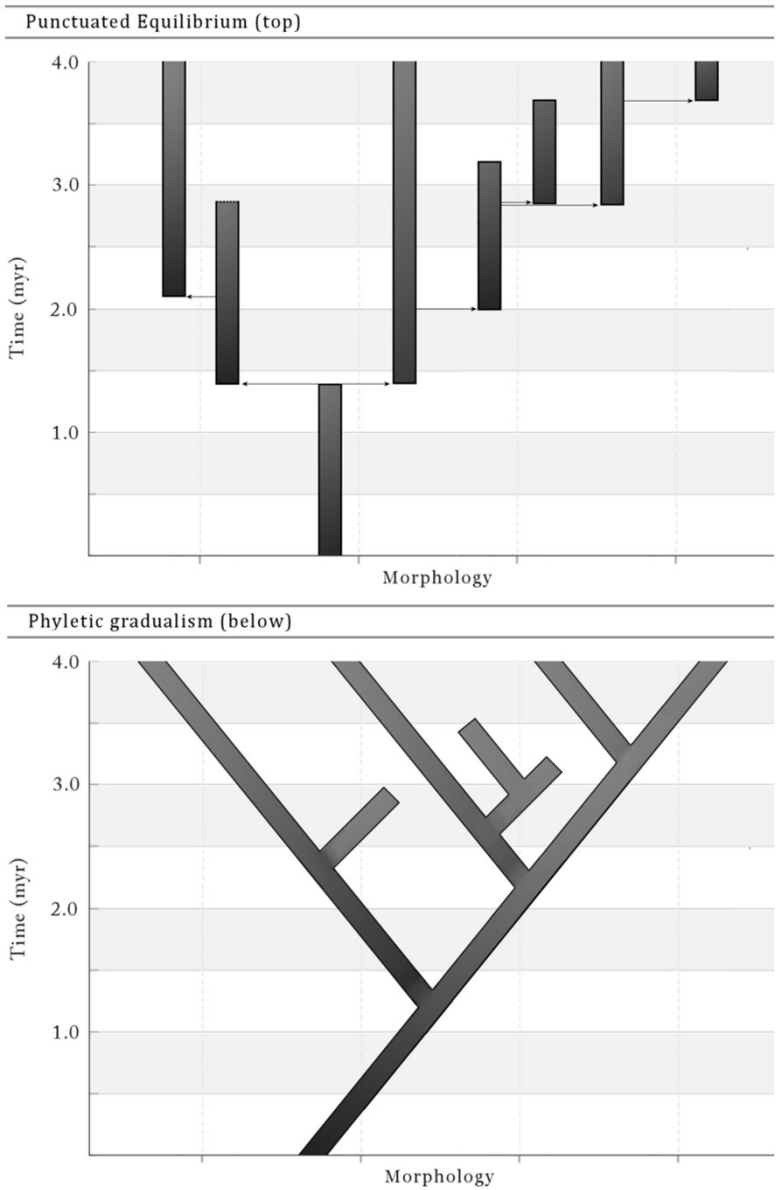
Figure 1.11
George Gaylord Simpson (1902–84) with a baby guanaco in central Patagonia in 1930.

George Gaylord Simpson argued that the evidence in the fossil record was consistent with the population genetic mechanisms of the modern synthesis.



Stephen Jay Gould

"Punctuated equilibrium"



The Neo-Darwinian Synthesis

The neo-Darwinian school of thought insisted that **populations rather than individuals should be the focus of** evolution.

Gene frequencies are important, and not just the presence or absence of genes.

Fisher, Wright and Haldane developed this approach towards the understanding of evolution, which was now defined as a **change in gene frequencies.**

In the 1920s and 1930s, this led to the emergence of the field of **population genetics**.

The Neo-Darwinian Synthesis

The population has the variability necessary to explain evolutionary genetic change through space and time, whereas an individual is extremely limited in these dimensions.

Differences among a population's genotypes enable different reproductive rates among them, whereas an individual's genotype is constant from birth to death.

Evolutionary changes, thus, depend on differentiation among genotypes: **populations evolve, individuals don't**.

The Neo-Darwinian Synthesis

Mutations occur randomly --- furnish the fuel for evolution by introducing genetic variability.

Evolution is thus an ongoing process --- random mutation introduces genes whose frequencies change through time, with natural selection usually considered as the most important, although not the only cause for such change.

The accumulation of gene frequency differences over time → more pronounced differences among populations in geographically different localities.

The Neo-Darwinian Synthesis

Due to reproductive barriers between populations, gene exchange is prevented, and speciation occurs.

Thus evolution cycles continuously from mutation to variation to selection to adaptation to organismic diversity.

The Neo-Darwinian synthesis thus provided an ideological framework to understand this continuum.

This initiated a spate of research that brought together researchers from very different disciplines of biology.